Lecture 19

Tuesday, 11 April 2023 10:35 AM

Start project!, Send email about bonus lecture

Alternate Scaling Relationships

Traditionally: correlate binding of 2 adsorbates across many surfaces

↳ Correlate binding energies of many adsorbates on a single surface.
(cf group contribution methods)

Bond-order parameter: $\frac{\chi_{max} - \chi}{\chi_{max}}$ → Use this to correlate many adsorbates:

$$BE_{\text{(``any'' adsorbate)}} = \sum_{\substack{i \text{ atoms} \\ \text{that bind} \\ \text{covalently} \\ \text{to surface}}} P_i \left(\frac{\chi_{max} - \chi}{\chi_{max}} \right) + \text{const.}$$

P_i: parameters fit to DFT data (different for each surface)

} Simple regression analysis

w/ a small no. of DFT calcs, we can estimate BEs of many species

↳ Can add higher order corrections: $BE = \sum_i P_i \left(\frac{\chi_{max} - \chi}{\chi_{max}} \right) + \sum_{\substack{i,j \\ \text{nearest} \\ \text{neighbor} \\ \text{atoms}}} P_{ij} \left(\frac{\chi_{max,i} - \chi_i}{\chi_{max,i}} \right) \left(\frac{\chi_{max,j} - \chi_j}{\chi_{max,j}} \right)$

P_{ij}: interactions b/w atoms

Classic Example: Scaling/BEP analysis (Mavrikakis et al., JACS (2004))

Consider ethanol partial oxidation on Transition Metal surfaces

$$C_2H_5OH_{(g)} + * \rightleftharpoons C_2H_5OH^*$$

$$C_2H_5OH^* + * \overset{C\text{-}H}{\rightleftharpoons} C_2H_4OH^* + H^*$$

$$\overset{O\text{-}H}{\searrow\!\nearrow} \quad C_2H_6O^* + H^* \overset{C\text{-}H}{\rightleftharpoons} C_2H_4O^* + H^*$$

$$C_2H_4OH^* + * \overset{C\text{-}H}{\rightleftharpoons} C_2H_3OH^* + H^*$$

$$\overset{O\text{-}H}{\searrow\!\nearrow} \quad C_2H_4O^* + H^*$$

⋮

More dehydration can occur, down to CCO* (never achieved in practice)

Intermediates:

$$C_2H_3OH^* + * \overset{C\text{-}C}{\rightleftharpoons} CH^* + CH_2OH^*$$

$$\overset{C\text{-}C}{\rightleftharpoons} CH_2^* + CHOH^*$$

Similar steps for other intermediates

$\overset{C\text{-}C}{\rightleftharpoons} CH_3^* + COH^*$

$\overset{C\text{-}O}{\rightleftharpoons} C_2H_3^* + OH^*$

To get maximum H_2 yield, suppress C-O bond scission (usually gives alkanes + H_2O)

How can one analyze such a complex mechanism?

Surface adsorption energies and surface coverages

↳ starting w/ a complete database of binding energies on Ru(0001), use scaling relationship

$$\Delta E - \Delta E_{Ru} = \sum_i \left(\frac{\chi_{max,i} - \chi_i}{\chi_{max,i}} \right) (\Delta E_i - \Delta E_{i/Ru}) \quad ; \quad i = C, O \text{ atoms}$$

↳ From these values, estimate θ's of all intermediates on all surface by assuming QE w/ appropriate gas species:

$$C_xH_y^* \rightleftharpoons C_3H_6(g) + \frac{y}{2} H_2(g) + *$$
$$C_xH_yO_z^* \rightleftharpoons C_2H_5OH(g) + \frac{z}{2} H_2(g) + *$$
$$CH_x^* \rightleftharpoons CH_4(g) + \frac{x}{2} H_2(g) + *$$

} Assume not rate limiting

⇓

For all metals except Cu, Ag, Au ; CO^* is the MASI

These had * as the MASI

} This is an approximation as well but assuming that C-H bond breaking is QE?

→ Since CO^* is at high coverage, account for interactions of species w/ CO^*

$$BE_{CO}(\theta_{CO}) = -1.78 + 0.0065\, e^{4.79\theta_{CO}} + 0.314\, \theta_{CO}\, e^{4.79\theta_{CO}}$$ } Pt(111) [PW91, differential BE = $\frac{dE}{d\theta}$]

↳ Crude first approximation: assume θ_{CO}-dependent terms for all adsorbates on all metals.

(b) Kinetics: + Use a BEP relationship to estimate barriers (energies of TS) for all possible reactions w/ C-C or C-O cleavage.

2 approaches: (i) correct reactant / product BE for θ_{CO}, then apply BEP

(ii) Apply BEP w/o θ_{CO}, then correct TS & reactant/product separately for θ_{CO}

} Barriers

+ Use coverages from thermo analysis

+ Calculate rates on each metal according to most stable TS for C-C & C-O bond breaking

$$R = 10^{13}\,s^{-1}\,\theta_{TS}\,\theta_{*}$$

$$\left(\approx \frac{k_B T}{h} K_{TS}\,\theta_{TS}\,\theta_{*}\right)$$

(c) Steps / Structure Sensitivity

+ general approach is to calculate ΔE to dissociate CO^* on steps

→ if $\Delta E < 0$, then steps will be poisoned w/ C^* ↔ do not contribute

→ if $\Delta E > 0$, steps must be considered

Con: not stated which is better

→ Redo calcs on steps